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TRACKING NUMBER	**CS-SA12-26014**
COHORT	CYBERSHUJAA COHORT 1-2026
TASK	PYTHON BASICS REPORT

Contents

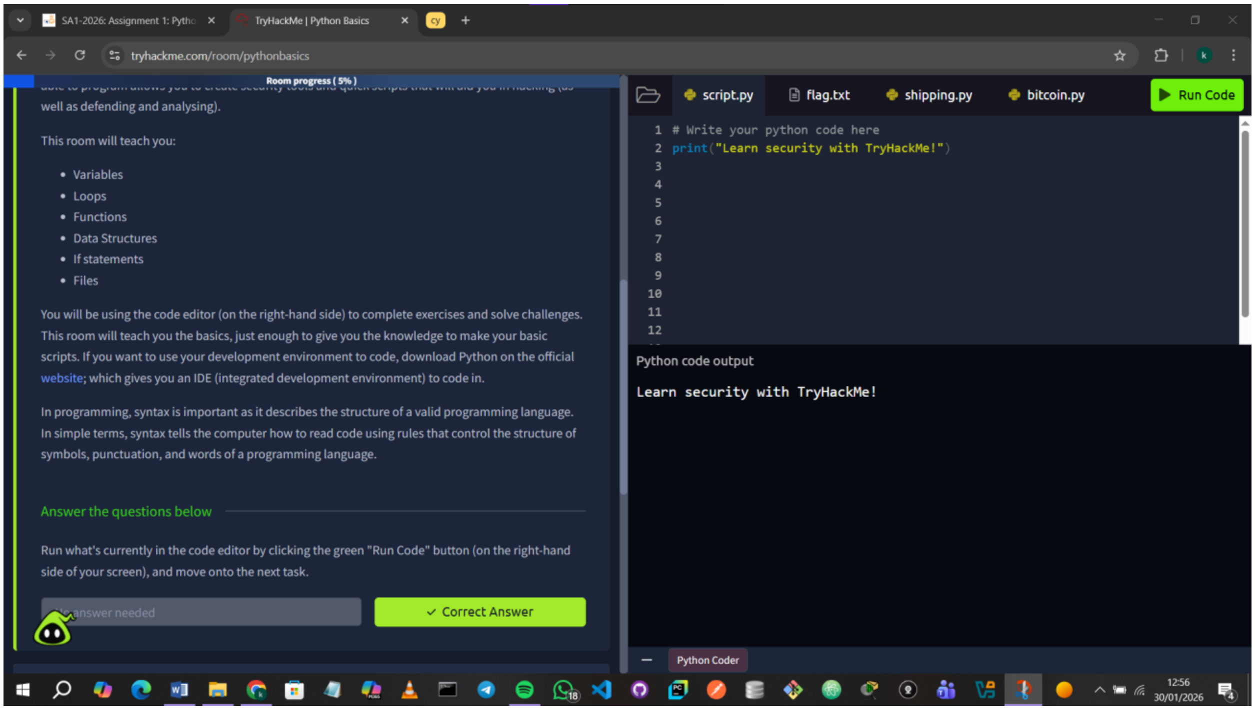
PYTHON BASICS REPORT

Introduction

In this module, I worked hands-on with Python and strengthened my scripting skills for cybersecurity tasks. I explored variables, data types, loops, functions, conditionals, file handling, and library imports, gaining practical experience in automating tasks and developing small tools. This module reinforced my existing Python proficiency while giving me new insights into applying it in a security context.

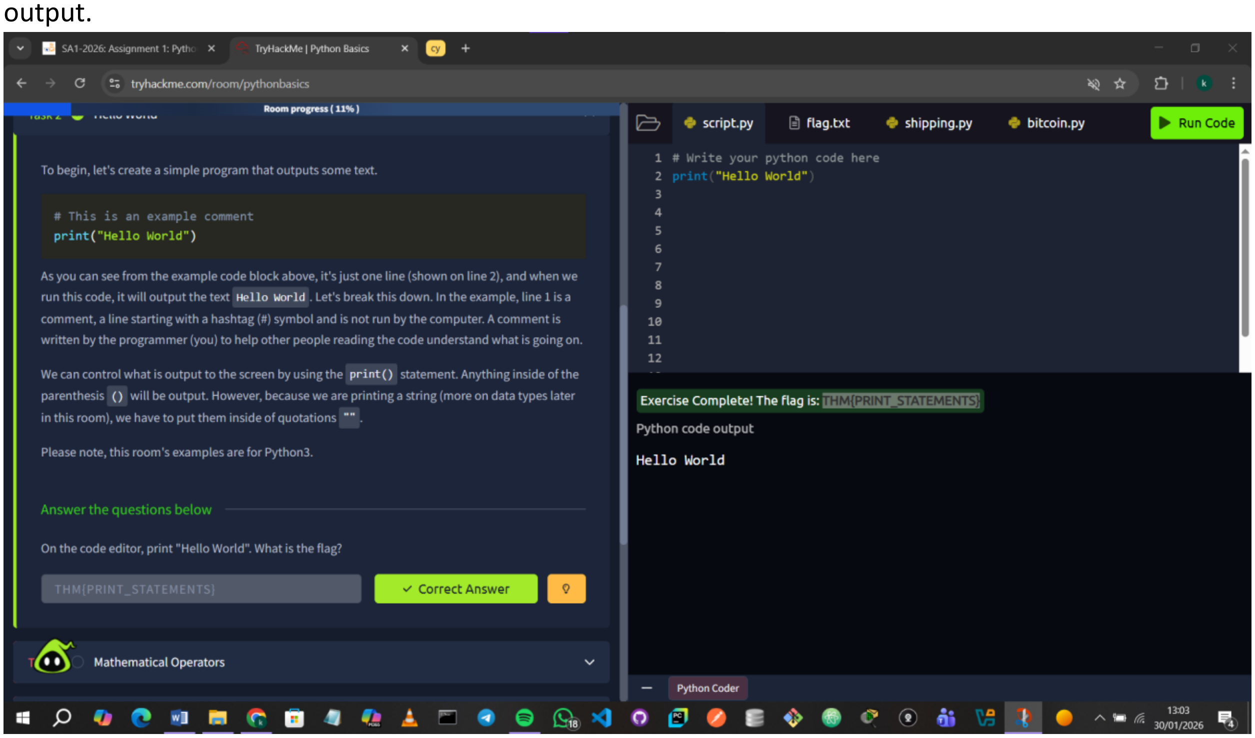
1.0 Introduction to Python

In this section, I learnt the fundamentals of Python and how it can be used for scripting in cybersecurity.



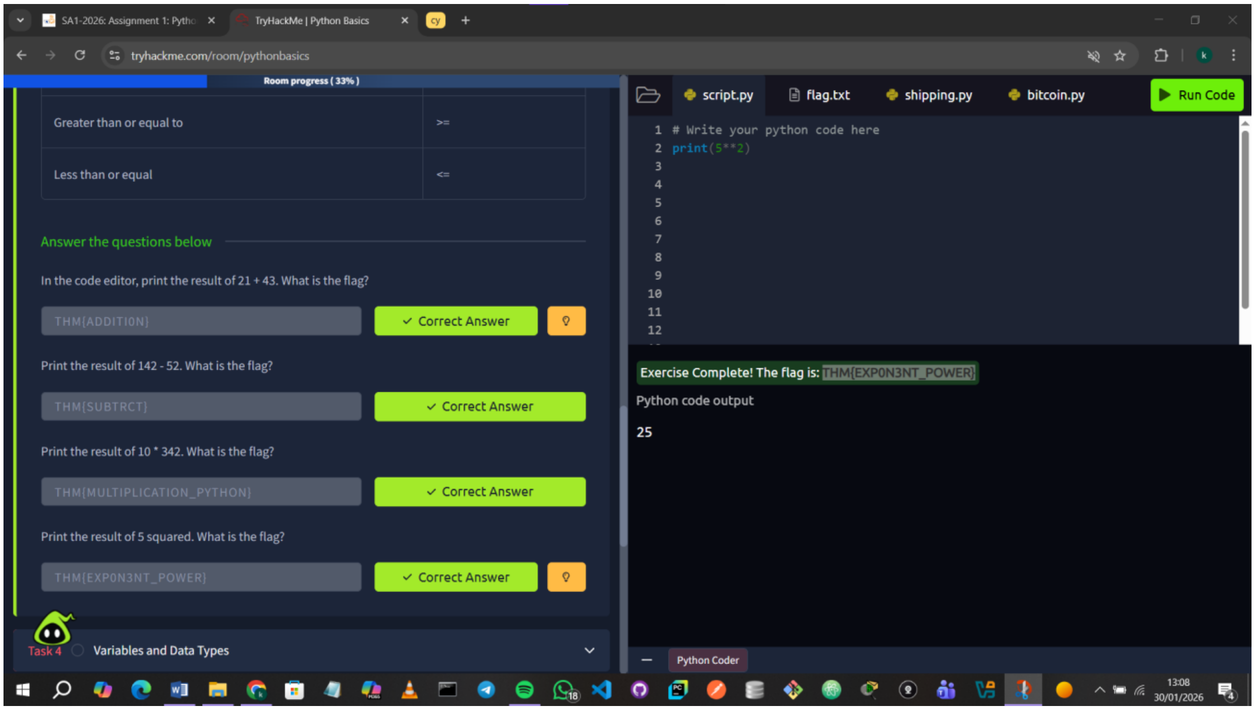
2.0 Hello World

In this section, I learnt how to write a simple Python program using the print() function to display output.



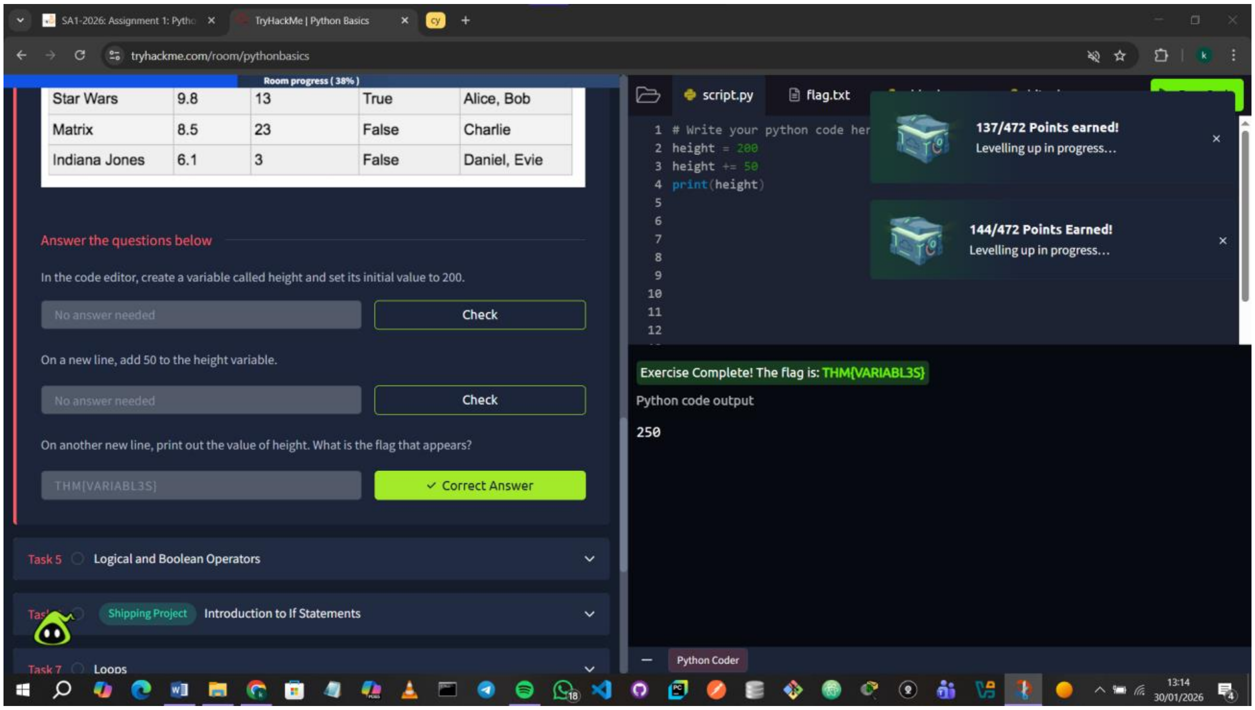
3.0 Mathematical operations

In this section, I learnt how Python uses mathematical operators for calculations like addition, subtraction, multiplication, and division. I also learnt about comparison operators, which are used to compare values and are important for making decisions in programs.



4.0 Variables and Data Types

In this section, I learnt how variables are used to store and update data in a program. I also learnt about different data types in Python, such as strings, numbers, Booleans, and lists, and how they define the kind of data a variable can hold.



5.0 Logical and Boolean Operations

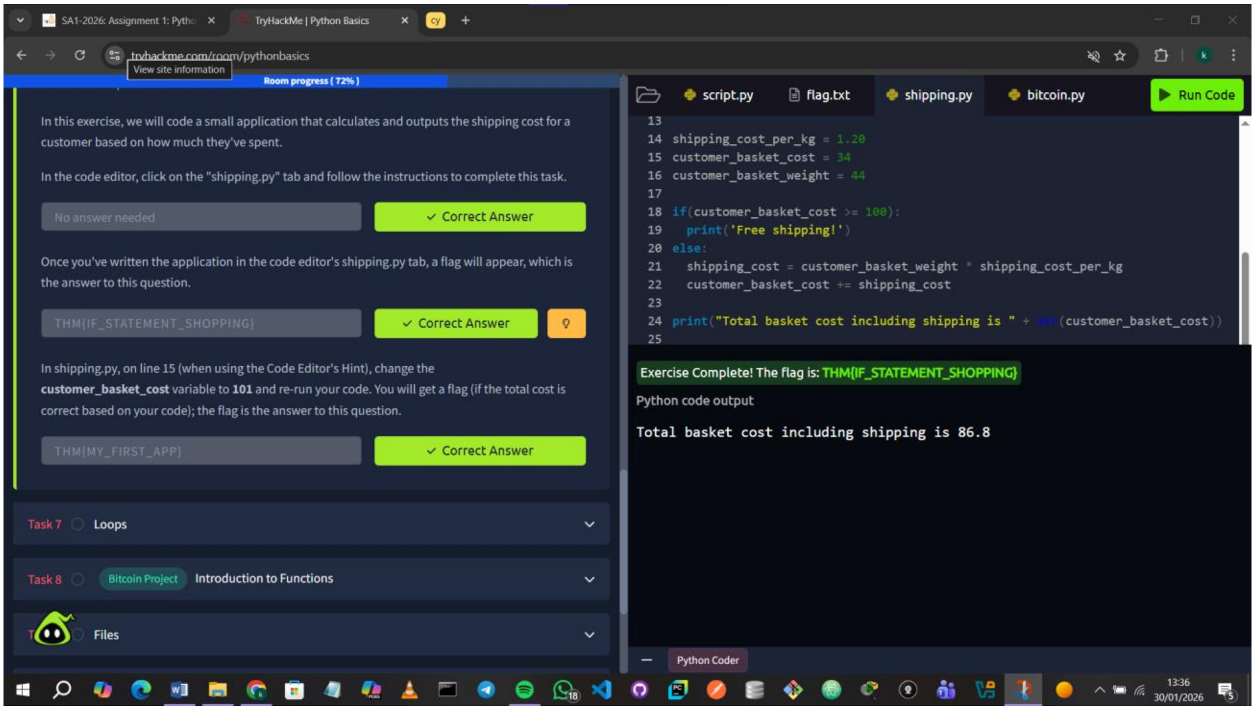
In this section, I learnt how logical and Boolean operators are used to compare values and test conditions in Python. I also learnt how operators like **AND**, **OR**, and **NOT** help combine conditions and control how a program makes decisions based on true or false results.

6.0 Introduction to If Statements

In this section, I learnt how **if statements** allow a program to make decisions based on conditions. I learnt how different blocks of code run depending on whether a condition is true or false, and how indentation and colons define what code belongs inside the if or else statements.

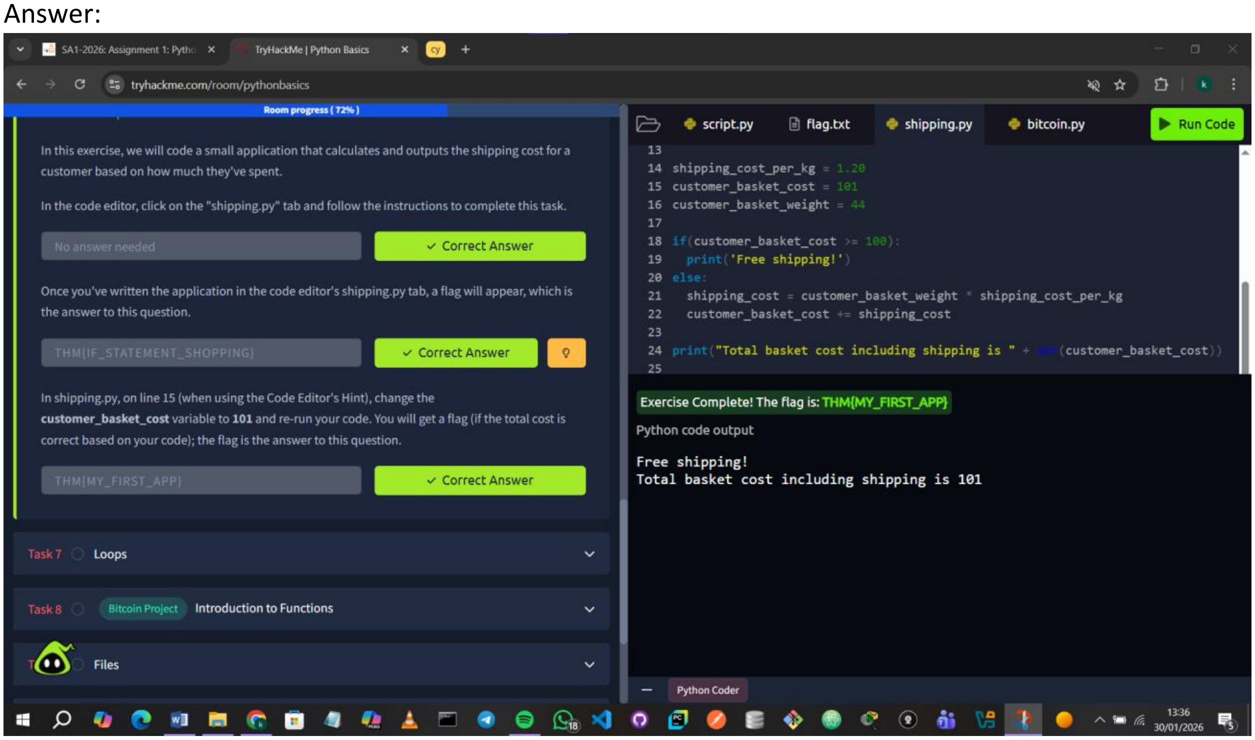
Question: Once you've written the application in the code editor's shipping.py tab, a flag will appear, which is the answer to this question.

Answer: I used if statements to decide whether or not the customer had to pay for shipping.



Question: In shipping.py, on line 15 (when using the Code Editor's Hint), change the customer basket cost variable to 101 and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.

Answer:

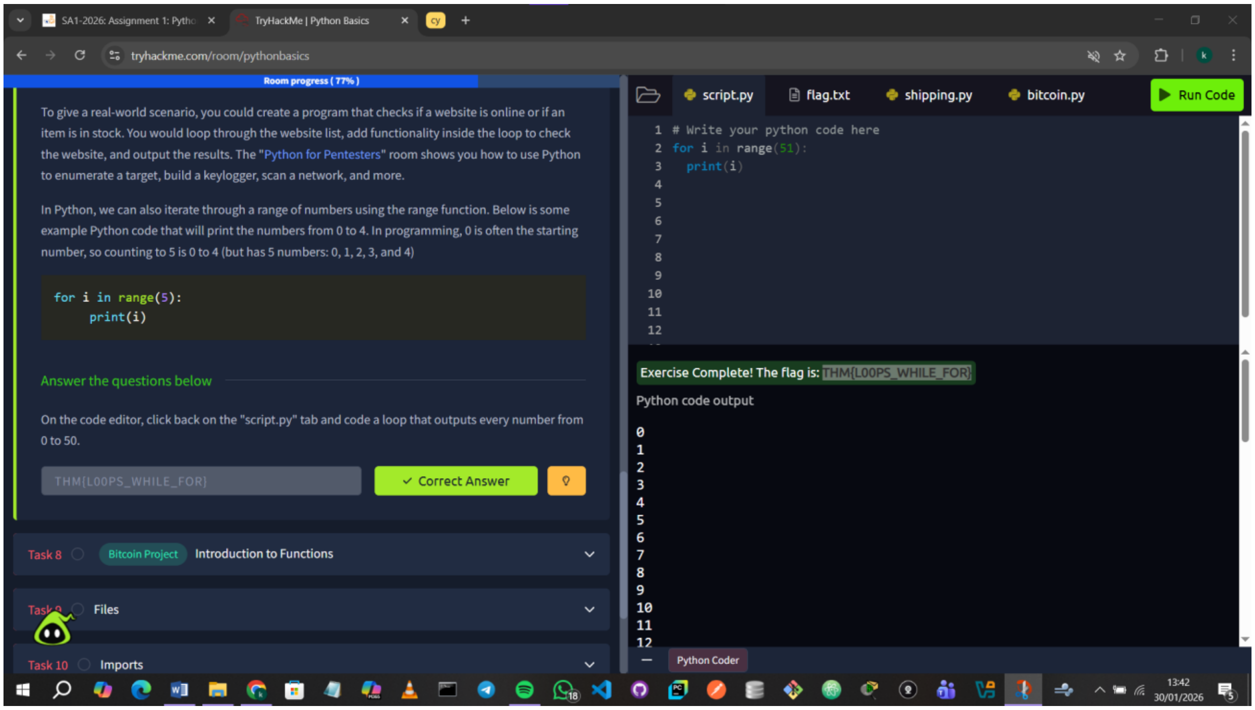


7.0 Loops

In this section, I learnt how **loops** let a program repeat actions automatically. I learnt how **while loops** run based on a condition and how **for loops** iterate through lists or ranges, which is useful for automating tasks like checking multiple items or targets.

Question: code a loop that outputs every number from 0 to 50.

Answer: I used **range(51)**

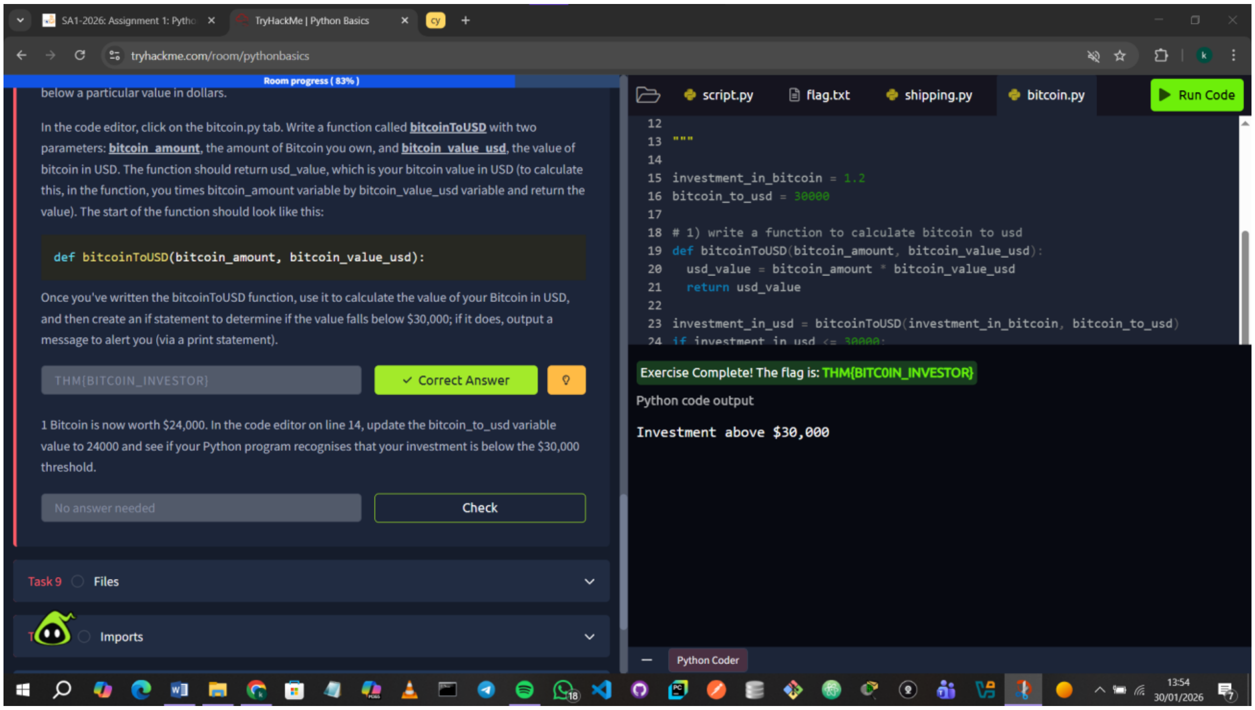


8.0 Introduction to Functions

In this section, I learnt that **functions** are reusable blocks of code that help avoid repetition.

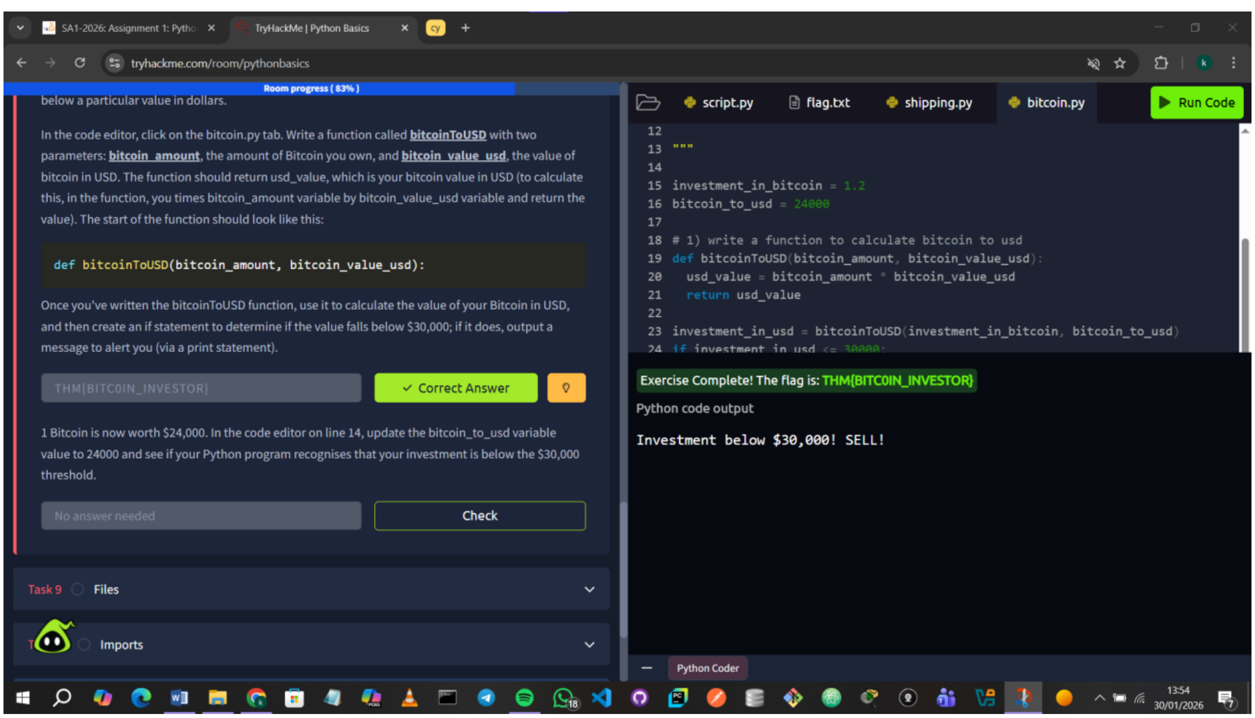
Question: Once you've written the bitcoinToUSD function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below $30,000; if it does, output a message to alert you (via a print statement).

Answer: I made the function then applied if statements to decide what should be printed out.



Question: 1 Bitcoin is now worth $24,000. In the code editor on line 14, update the bitcoin to usd variable value to 24000 and see if your Python program recognises that your investment is below the $30,000 threshold.

Answer:

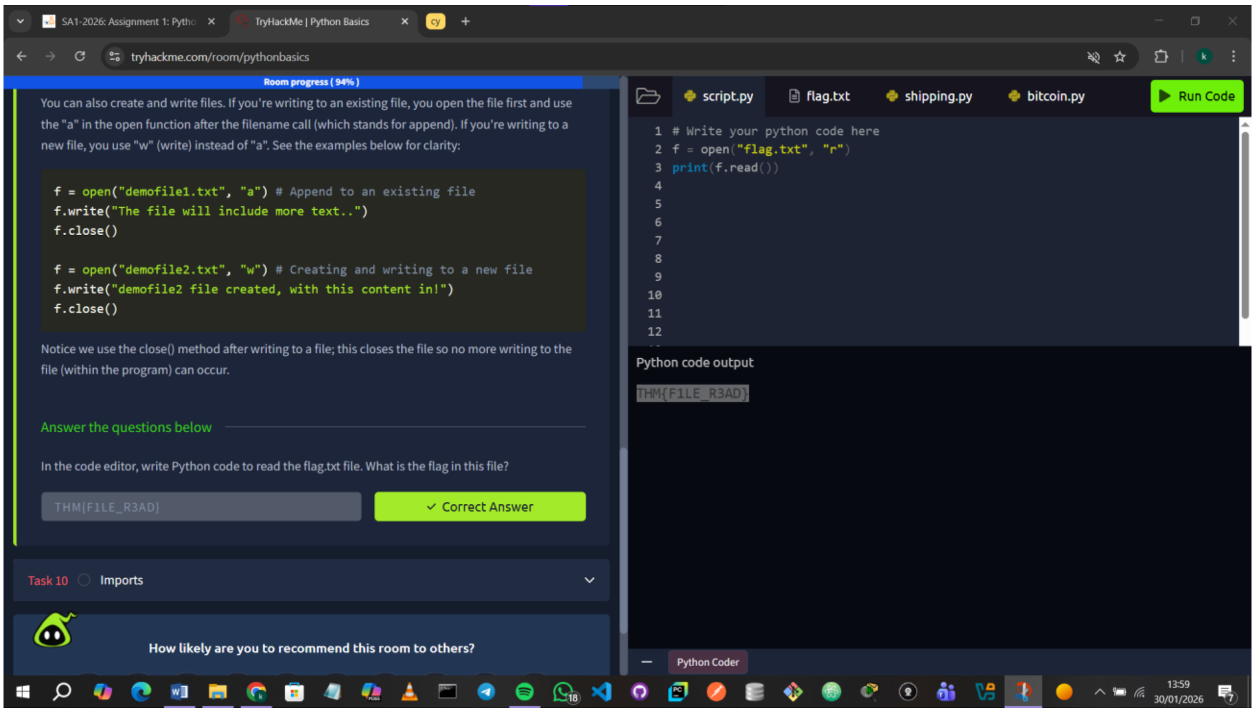


9.0 Files

In this section, I learnt how to **read from and write to files in Python**. I learnt how to open files in different modes (r, a, w), read their contents, write or append data, and properly close files to save changes.

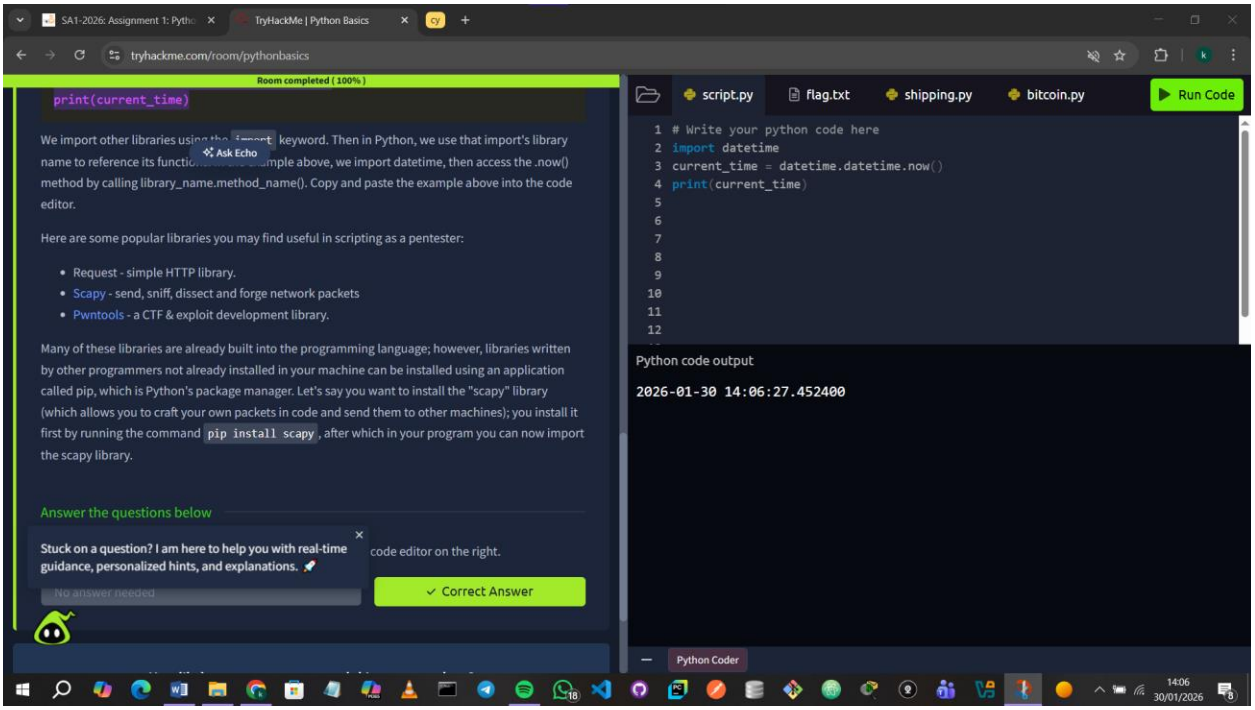
Question: In the code editor, write Python code to read the flag.txt file. What is the flag in this file?

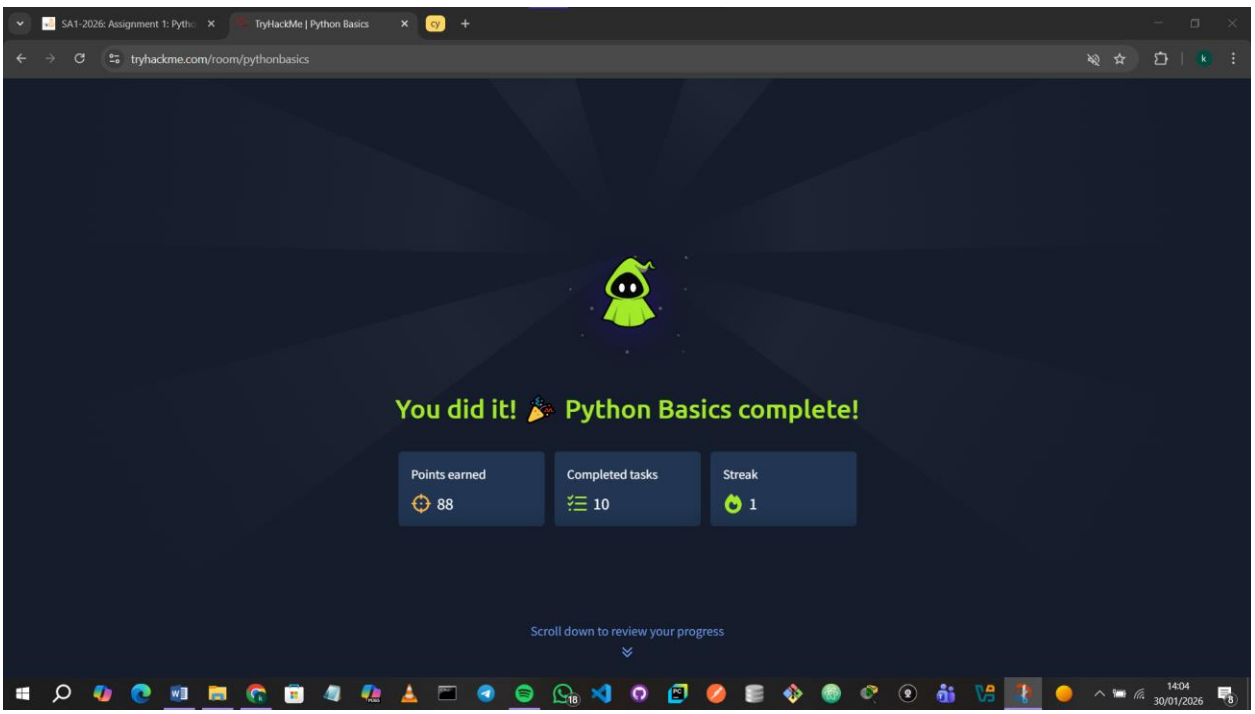
Answer: I used **open** to open the file and **r** to read it then printed it out.



10.0 Imports

In this section, I learnt how to **import and use Python libraries**.





Conclusion

This module covered the essentials of Python, from variables and loops to functions, file handling, and libraries. As someone already proficient in Python, it was a great refresher and helped reinforce best practices for scripting and automation.